



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Friday, 11 September 2026, 4:13 PM**State** Finished**Completed on** Friday, 11 September 2026, 4:13 PM**Time taken** 32 secs**Marks** 1.00/1.00**Grade** 4.00 out of 4.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main() {
5      int n;
6      if (scanf("%d", &n) != 1) return 0;
7
8      int *nums = (int *)malloc(n * sizeof(int));
9      for (int i = 0; i < n; i++) {
10         if (scanf("%d", &nums[i]) != 1) return 0;
11     }
12
13     int tortoise = nums[0];
14     int hare = nums[0];
15
16     do {
17         tortoise = nums[tortoise];
18         hare = nums[nums[hare]];
19     } while (tortoise != hare);
20
21     tortoise = nums[0];
22     while (tortoise != hare) {
23         tortoise = nums[tortoise];
24         hare = nums[hare];
25     }
26
27     printf("%d\n", tortoise);
28
29     free(nums);
30     return 0;
31 }
32

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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